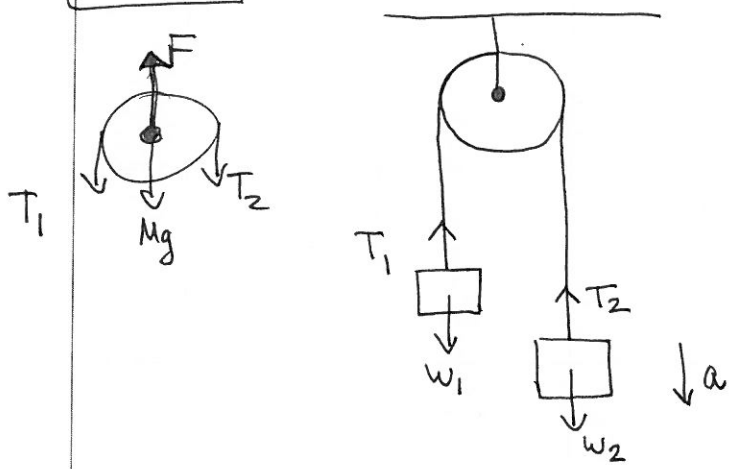
P10.67



$w_1 = 75.0$ N

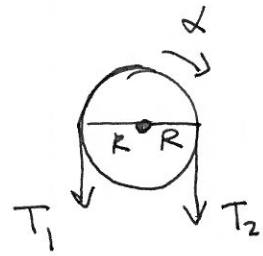
$w_2 = 125$ N

pulley 50.0 N radius $R = 0.300$ m

$$-w_2 + T_2 = -m_2 a$$

$$T_1 - w_1 = m_1 a$$

$$\tau = -T_2 R + T_1 R = -I\alpha$$



no slipping $\Rightarrow$ $R\alpha = a$

$$I = \frac{1}{2} M R^2$$

Note: $T_2 > T_1$ so that pulley experience torque.

$$\Rightarrow T_2 - T_1 = \frac{1}{2} M R \alpha = \frac{1}{2} M a$$

$$T_2 = w_2 - m_2 a$$

$$T_1 = w_1 + m_1 a$$

$$\Rightarrow + w_2 - m_2 a - w_1 - m_1 a = \frac{1}{2} M a$$

$$w_2 - w_1 = \left(m_1 + m_2 + \frac{1}{2} M\right) a \Rightarrow a = \frac{w_2 - w_1}{m_1 g + m_2 g + \frac{1}{2} M g} g$$

$$a = \frac{w_2 - w_1}{w_1 + w_2 + \frac{1}{2} M g} g = \frac{125 - 75}{125 + 75 + \frac{1}{2}(50)} g = \frac{50}{225} g$$

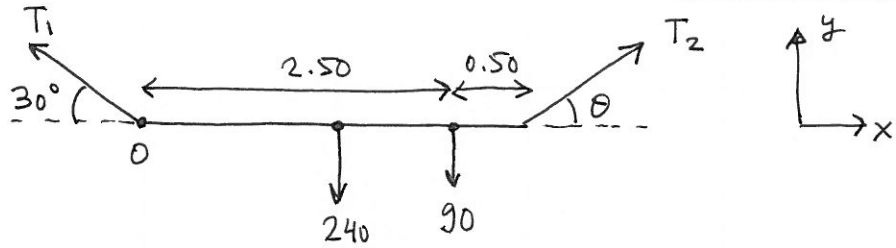
Acceleration $a = 0.222 g$

$$T_2 = w_2 - 0.222\, m_2 g = 0.778\, w_2 = 97.25 \text{ N}$$

$$T_1 = w_1 + 0.222\, m_1 g = 1.222\, w_1 = 91.65 \text{ N}$$

$$\Rightarrow F = Mg + T_1 + T_2 = 239 \text{ N}$$

P11.19



$$\begin{cases} T_1 \sin 30° + T_2 \sin\theta - 240 - 90 = 0 \\ -T_1 \cos 30° + T_2 \cos\theta = 0 \\ \tau_o = -240(1.5) - 90(2.5) + T_2 \sin\theta (3.0) = 0 \end{cases}$$

$$\Rightarrow T_2 \sin\theta = 195$$

$$\Rightarrow T_1 \sin 30° = 240 + 90 - T_2 \sin\theta = 135$$

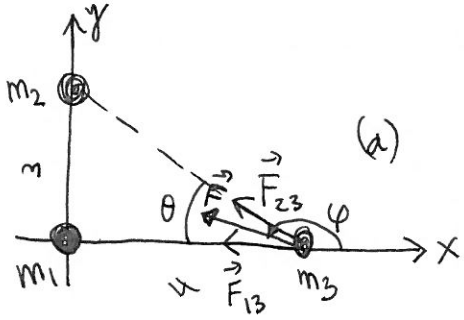
$$T_1 = \frac{135}{\sin 30°} = 270\ N$$

$$T_2 \cos\theta = T_1 \cos 30° = 234$$

$$\Rightarrow \frac{T_2 \sin\theta}{T_2 \cos\theta} = \frac{195}{234} \Rightarrow \tan\theta = 0.833 \Rightarrow \theta = 40.0°$$

$$T_2 = \frac{195}{\sin 40°} = 303\ N \ ; \ T_1 = 270\ N; \ \theta = 40.0°$$

P12.50



(a)

$m_1 = 60.0\ kg$, $m_2 = 80.0\ kg$, $m_3 = 0.500\ kg$

$\tan\theta = \frac{3}{4} = 0.75$; $\cos\theta = \frac{4}{5}$, $\sin\theta = \frac{3}{5}$

$\vec{F}_{13} = -|F_{13}|\,\hat{i}$

$r_{13} = 4.00\ m$

$r_{23} = 5.00\ m$

$\vec{F}_{23} = -|F_{23}|\cos\theta\,\hat{i} + |F_{23}|\sin\theta\,\hat{j}$

$$|F_{13}| = \frac{G m_1 m_3}{r_{13}^2} = 1.251 \cdot 10^{-10}\ N$$

$$|F_{23}| = \frac{G m_2 m_3}{r_{23}^2} = 1.068 \cdot 10^{-10}\ N$$

Resultant force $\vec{F} = \vec{F}_{13} + \vec{F}_{23} = -\left(|F_{13}| + |F_{23}| \cos\theta\right)\hat{\imath}$

$$+ |F_{23}| \sin\theta\, \hat{\jmath} =$$

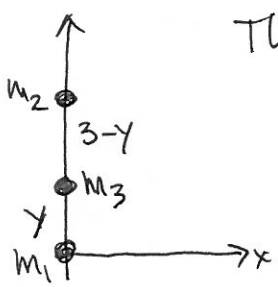
$$= -2.105 \cdot 10^{-10}\ N\ \hat{\imath} + 6.408 \cdot 10^{-10}\ N\ \hat{\jmath}$$

Magnitude $|\vec{F}| = \sqrt{F_x^2 + F_y^2} = 2.20 \cdot 10^{-10}\ N$

Direction $\tan\varphi = \frac{F_y}{F_x} = \frac{0.641 \cdot 10^{-10}}{-2.105 \cdot 10^{-10}} \Rightarrow \varphi = 163^\circ$

(measured from positive $\hat{x}$)

(b) To have net grav. force $\vec{F} = 0$ the ball m_3 must be placed between m_1 and m_2:



Then
$$\left.\begin{aligned} \vec{F}_{13} &= -\frac{G m_1 m_3}{y^2}\,\hat{\jmath} \\ \vec{F}_{23} &= +\frac{G m_2 m_3}{(3-y)^2}\,\hat{\jmath} \end{aligned}\right\} \vec{F} = \vec{F}_{13} + \vec{F}_{23} = 0$$

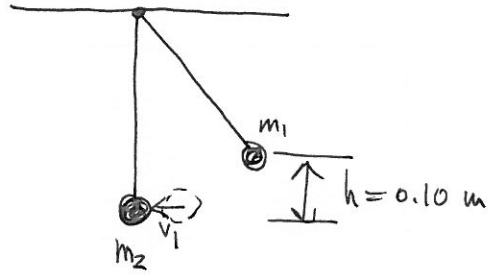
$$\Rightarrow \frac{m_2}{(3-y)^2} - \frac{m_1}{y^2} = 0$$

$$\left(\frac{3-y}{y}\right)^2 = \frac{m_2}{m_1} = \frac{80}{60} = \frac{4}{3}$$

$$\frac{3-y}{y} = \sqrt{\frac{4}{3}}, \quad 3 - y = \sqrt{\frac{4}{3}}\, y, \quad 3 = \left(1 + \sqrt{\frac{4}{3}}\right) y \Rightarrow y = \frac{3}{1 + \sqrt{\frac{4}{3}}} = 1.39\ m$$

P13.89 Inelastic collision

(1)



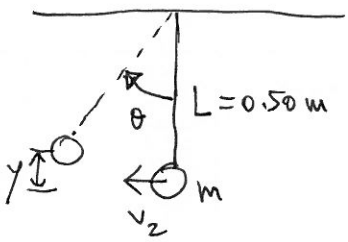
$m_1 = 2.00$ kg
$m_2 = 3.00$ kg

Momentum: $m_1 v_1 = (m_1 + m_2) V_2 \Rightarrow V_2 = \frac{m_1}{m_1 + m_2} V_1 = \frac{2}{5} V_1 = 0.4 V_1$

Speed v_1 just before collision: $m_1 g h = \frac{1}{2} m_1 v_1^2 \Rightarrow v_1 = \sqrt{2gh} = 1.40$ m/s

$\Rightarrow V_2 = 0.56$ m/s

(2) Now the combined object $m = m_1 + m_2 = 5.00$ kg moves as one:



rises up to $mgy = \frac{1}{2} m v_2^2$

$y = \frac{v_2^2}{2g} = 0.016$ m

Angular displacement θ:

$y = L - L\cos\theta$, $\cos\theta = \frac{L-y}{L} = 1 - \frac{0.016}{0.50} = 0.968$

$\Rightarrow \theta = \cos^{-1}(0.968) = 14.5^\circ$

$T = 2\pi\sqrt{\frac{L}{g}} \Rightarrow f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{g}{L}} = 0.705\ s^{-1}$